Stage 10/02/2026

① $R_0 \to R$

$\xi^\alpha \to x^\mu$

$$\frac{d^2\xi^\alpha}{d\tau^2} = 0$$

$$\Leftrightarrow \frac{d^2x^\sigma}{d\tau^2} + \frac{\partial x^\sigma}{\partial \xi^\alpha}\frac{\partial \xi^\alpha}{\partial x^\mu \partial x^\nu}\frac{dx^\mu}{\partial \tau}\frac{dx^\nu}{d\tau} = 0$$

$$\frac{d^2\xi^\alpha}{d\tau^2} = \frac{d}{d\tau}\left(\frac{d\xi^\alpha}{d\tau}\right) = 0$$

$$\Leftrightarrow \frac{d}{d\tau}\left(\frac{d\xi^\alpha}{dx^\mu} \times \frac{dx^\mu}{d\tau}\right) = 0$$

$$\Leftrightarrow 0 = \frac{d\xi^\alpha}{dx^\mu}\frac{d^2x^\mu}{d\tau} + \frac{dx^\mu}{d\tau} \times \frac{d^2\xi^\alpha}{dx^\mu dx^\nu} \times \frac{dx^\nu}{d\tau}$$

$$\Leftrightarrow 0 = \frac{d\xi^\alpha}{dx^\mu}\frac{dx^\rho}{d\xi^\alpha}\frac{d^2x^\mu}{d\tau^2} + \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}\frac{d^2\xi^\alpha}{dx^\mu dx^\nu}\frac{dx^\rho}{d\xi^\alpha}$$

$$= \frac{d^2x^\rho}{d\tau^2} + \left\{ {\rho \atop \mu\nu} \right\}\frac{\partial x^\nu}{\partial \tau}\frac{\partial x^\mu}{\partial \tau}$$

② $$g_{\mu\nu} = \frac{\partial \xi^\alpha}{\partial x^\mu}\frac{\partial \xi^\beta}{\partial x^\nu}\eta_{\alpha\beta}$$

$$\partial_\Delta g_{\mu\nu} = \frac{\partial}{\partial x^\Delta}\left(\frac{\partial \xi^\alpha}{\partial x^\mu}\frac{\partial \xi^\beta}{\partial x^\nu}\eta_{\alpha\beta}\right)$$

$$= \frac{\partial^2 \xi^\alpha}{\partial x^\Delta \partial x^\mu}\frac{\partial \xi^\beta}{\partial x^\nu}\eta_{\alpha\beta} + \frac{\partial^2 \xi^\beta}{\partial x^\Delta \partial x^\mu}\frac{\partial \xi^\alpha}{\partial x^\nu}\eta_{\alpha\beta}$$

$$= \underbrace{\frac{\partial x^\rho}{\partial y^\alpha} \frac{\partial^2 y^\alpha}{\partial x^\Delta \partial x^\mu}}_{\left\{ {\rho \atop \Delta\ \mu} \right\}} \underbrace{\frac{\partial y^\beta}{\partial x^\nu} \times \frac{\partial y^\alpha}{\partial x^\rho} \eta_{\alpha\beta}}_{g_{\rho\nu}} + \dots$$

$$= \left\{ {\rho \atop \Delta\ \mu} \right\} g_{\rho\nu} + \left\{ {\rho \atop \Delta\ \nu} \right\} g_{\rho\mu}$$

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\beta\alpha} \left(\frac{\partial g_{\alpha\mu}}{\partial x^\nu} + \frac{\partial g_{\alpha\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \right)$$

$$\Gamma^\Delta_{\mu\ \nu} = \frac{1}{2} g^{\Delta\alpha} \left(\partial_\nu g_{\alpha\mu} + \partial_\mu g_{\alpha\nu} - \partial_\alpha g_{\mu\nu} \right)$$

$$= \frac{1}{2} g^{\Delta\alpha} \Big[\underbrace{\left\{ {\rho \atop \nu\ \mu} \right\} g_{\rho\alpha} + \left\{ {\rho \atop \nu\ \alpha} \right\} g_{\rho\mu}}$$

$$+ \underbrace{\left\{ {\rho \atop \mu\ \alpha} \right\} g_{\rho\nu} + \left\{ {\rho \atop \nu\ \mu} \right\} g_{\rho\alpha}}$$

$$- \underbrace{\left\{ {\rho \atop \alpha\ \mu} \right\} g_{\rho\nu} - \left\{ {\rho \atop \alpha\ \nu} \right\} g_{\rho\mu}} \Big]$$

$$= \frac{1}{2} g^{\Delta\alpha} \times 2 \left\{ {\rho \atop \nu\ \mu} \right\} g_{\rho\alpha}$$

$$= \left\{ {\rho \atop \nu\mu} \right\} g^{\Delta\alpha} g_{\rho\alpha}$$

$$= \left\{ {\Delta \atop \mu\ \nu} \right\}$$

(3) Loi de trafo

$$x \longrightarrow x'$$

$$\left\{ {\Delta \atop \mu\ \nu} \right\}' = \frac{\partial x'^{\Delta}}{\partial \xi^{\alpha}} \frac{\partial^2 \xi^{\alpha}}{\partial x'^{\mu} \partial x'^{\nu}}$$

$$= \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial \xi^{\alpha}} \times \frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial \xi^{\alpha}}{\partial x'^{\nu}} \right)$$

$$= \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial \xi^{\alpha}} \times \frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial x^{\beta}}{\partial x'^{\nu}} \frac{\partial \xi^{\alpha}}{\partial x^{\beta}} \right)$$

$$= \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial \xi^{\alpha}} \times \left[\frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial x^{\beta}}{\partial x'^{\nu}} \right) \frac{\partial \xi^{\alpha}}{\partial x^{\beta}} + \frac{\partial x^{\beta}}{\partial x'^{\nu}} \frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial \xi^{\alpha}}{\partial x^{\beta}} \right) \right]$$

$$= \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial \xi^{\alpha}} \times \frac{\partial^2 x^{\beta}}{\partial x'^{\mu} \partial x'^{\nu}} \frac{\partial \xi^{\alpha}}{\partial x^{\beta}} + \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial \xi^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\mu}} \frac{\partial x^{\gamma}}{\partial x'^{\nu}} \frac{\partial^2 \xi^{\alpha}}{\partial x^{\beta} \partial x^{\gamma}}$$

$$= \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \underbrace{\left\{ {\rho \atop \beta \quad \gamma} \right\}} \frac{\partial x^{\beta}}{\partial x'^{\mu}} \frac{\partial x^{\gamma}}{\partial x'^{\nu}} + \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}}$$

$$\frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}} = \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial}{\partial x'^{\nu}} \left(\frac{\partial x^{\rho}}{\partial x'^{\mu}} \right)$$

$$\frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial^2 x^{\rho}}{\partial x'^{\mu} \partial x'^{\nu}} = \frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial x^{\rho}}{\partial x'^{\nu}} \right)$$

$$= \frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial x'^{\Delta}}{\partial x^{\rho}} \frac{\partial x^{\rho}}{\partial x'^{\nu}} \right) - \frac{\partial x^{\rho}}{\partial x'^{\nu}} \frac{\partial}{\partial x'^{\mu}} \left(\frac{\partial x'^{\Delta}}{\partial x^{\rho}} \right)$$

$$= 0 - \frac{\partial x^{\rho}}{\partial x'^{\nu}} \frac{\partial x^{\sigma}}{\partial x'^{\mu}} \frac{\partial^2 x'^{\Delta}}{\partial x^{\sigma} \partial x^{\rho}}$$

(4) $\nabla_{\mu} V^{\nu} = \partial_{\mu} V^{\nu} + \Gamma^{\nu}_{\mu \rho} V^{\rho}$

$$\partial' \nabla'_{\mu} V'^{\nu} = \frac{\partial}{\partial x'^{\mu}} x'^{\nu} + \left\{ {\nu \atop \mu \quad \rho} \right\}' x'^{\rho}$$

$$= \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial}{\partial x^\beta}\left(\frac{\partial x'^\nu}{\partial x^\beta} x^\beta\right) + \left(\frac{\partial x'^\nu}{\partial x^\Delta}\frac{\partial x^\sigma}{\partial x'^\mu}\frac{\partial x^\tau}{\partial x'^\rho}\begin{Bmatrix}\Delta\\ \tau\ \sigma\end{Bmatrix}\right.$$

$$\left. - \frac{\partial x^\Delta}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\rho}\frac{\partial^2 x'^\nu}{\partial x^\Delta \partial x^\sigma}\frac{\partial x'^\rho}{\partial x^\beta} x^\beta\right)$$

$$= \frac{\partial x'^\alpha}{\partial x'^\mu}\left(\frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial x^\beta}{\partial x'^\alpha}\frac{\partial x^\alpha}{\partial x'^\mu} x^\beta + \right.$$

$$+ \frac{\partial x'^\nu}{\partial x^\Delta}\frac{\partial x^\sigma}{\partial x'^\mu}\frac{\partial x^\tau}{\partial x'^\rho}\frac{\partial x'^\rho}{\partial x^\beta}\begin{Bmatrix}\Delta\\ \tau\ \sigma\end{Bmatrix} x^\beta,$$

$$\left. - \frac{\partial x^\Delta}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\rho}\frac{\partial^2 x'^\nu}{\partial x^\Delta \partial x^\sigma}\frac{\partial x'^\rho}{\partial x^\beta} x^\beta\right)$$

$$= \frac{\partial x'^\alpha}{\partial x'^\mu}\frac{\partial x'^\nu}{\partial x^\beta} x^\beta - \frac{\partial x^\Delta}{\partial x'^\mu}\frac{\partial^2 x'^\nu}{\partial x^\Delta \partial x^\beta} x^\beta$$

$$+ \frac{\partial x'^\nu}{\partial x^\Delta}\frac{\partial x^\sigma}{\partial x'^\mu}\left(\partial_\sigma V^\Delta + \Gamma^\Delta_{\sigma\tau} x^\tau\right)$$

$$= \frac{\partial x'^\nu}{\partial x^\Delta}\frac{\partial x^\sigma}{\partial x'^\mu}\left(\partial_\sigma V^\Delta + \Gamma^\Delta_{\sigma\tau} V^\tau\right)$$

$$[\nabla_\mu, \nabla_\nu] V^\alpha = (\nabla_\mu \nabla_\nu - \nabla_\nu \nabla_\mu) V^\alpha$$

$$= \nabla_\mu\left(\partial_\nu V^\alpha - \Gamma^\alpha_{\nu\beta} V^\beta\right) - \nabla_\nu\left(\partial_\mu V^\alpha - \Gamma^\alpha_{\mu\beta} V^\beta\right)$$

$$\nabla_\mu(\partial_\nu V^\alpha - \Gamma^\alpha_{\nu\beta} V^\beta) - \nabla_\nu(\partial_\mu V^\alpha - \Gamma^\alpha_{\mu\beta} V^\beta)$$

$$= \partial_\mu \partial_\nu V^\alpha - \partial_\mu \Gamma^\alpha_{\nu\beta} V^\beta - \Gamma^\alpha_{\mu\beta} \partial_\nu V^\beta + \Gamma^\alpha_{\mu\gamma} \Gamma^\gamma_{\nu\beta} V^\beta$$

$$- \left(+ \partial_\nu \partial_\mu V^\alpha - \partial_\nu \Gamma^\alpha_{\mu\beta} V^\beta - \Gamma^\alpha_{\mu\beta} \partial_\nu V^\beta + \Gamma^\alpha_{\nu\gamma} \Gamma^\gamma_{\mu\beta} V^\beta \right)$$

$$= \left(\partial_\mu \Gamma^\alpha_{\nu\beta} V^\beta - \partial_\nu \Gamma^\alpha_{\mu\beta} V^\beta \right) + \left(\Gamma^\alpha_{\mu\beta} \Gamma^\beta_{\mu}$$